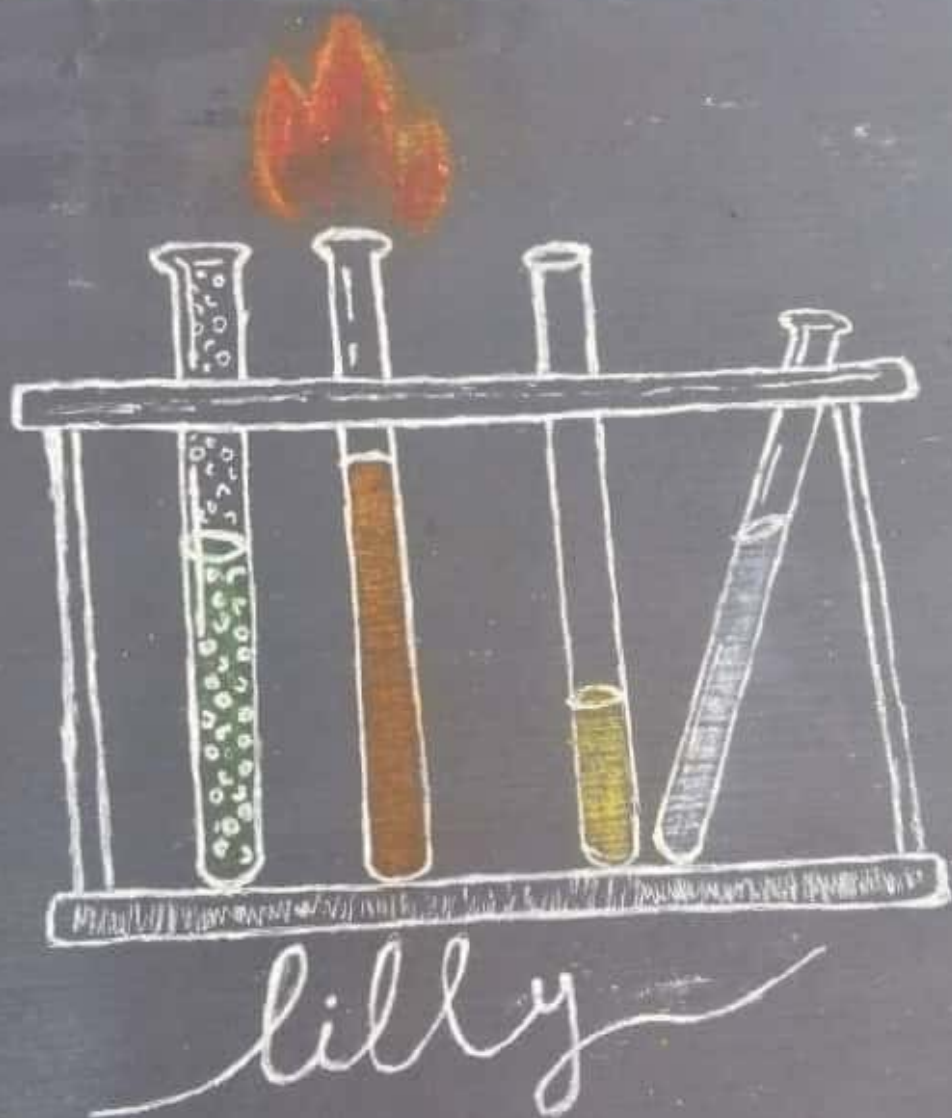
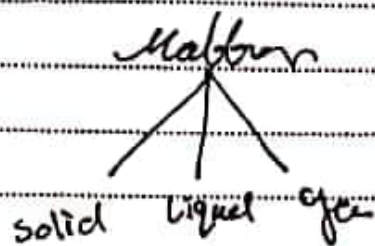


Chemistry



H2O_GROUP

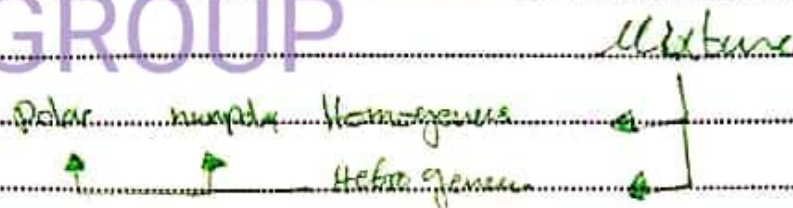
Subject: Matter and measurements



Intermolecular force depends on r

- distance between molecules
- The compact

H₂O GROUP

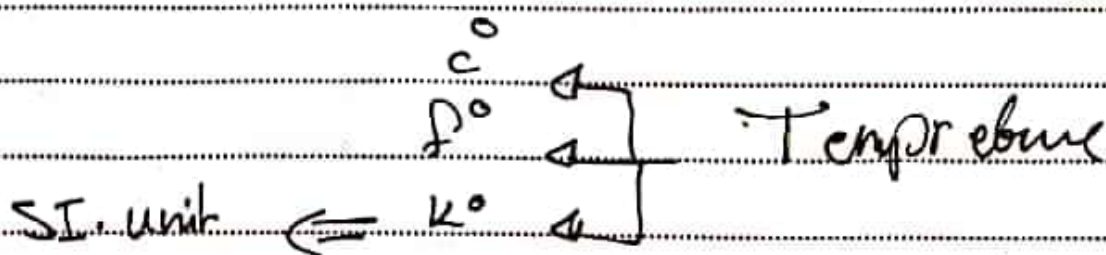


* Chemical properties :- "reactivity" gives new properties
- (Na) alone's Toxic but (NaCl) is essential for the human body

* Physical properties :- Same chemical bonds but a difference in the state (L, g, s)

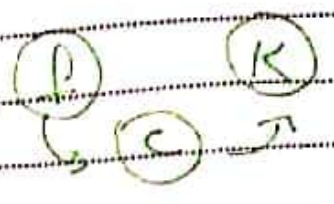
* Intensive :- mass independent (don't depend on mass)
[Temperature, pressure]

* Extensive :- mass dependent (depends on the mass)
[Volume, weight, molarity]



► Subject: Matter and measurement / S. P.

Law of $K = C^{\circ} + 273$
 $C^{\circ} = (F - 32) \times \frac{5}{9}$
 $F = (\frac{9}{5} \times C) + 32$



The absolute value of temp is [0 K]
(-40 C = -40 F)

→ $1g = 10^3 \text{ mg}$
 10^6 kg
 10^{12} pg

Density = $\frac{\text{mass}}{\text{Volume}}$

$1 \text{ m}^3 = 10^3 \text{ L} = 1000 \text{ L}$
 $1 \text{ L} = 1000 \text{ ml} = 10^3 \text{ ml}$
 $1 \text{ cm}^3 = 1 \text{ ml}$

H2O_GROUP

Significant Figures

The more sig. fig the more accurate

$1.03 \times 10^6 \Rightarrow 3 \text{ s.f.}$ [don't count the power]

$0.00025 \Rightarrow 2 \text{ s.f.}$ [don't count the zeros]

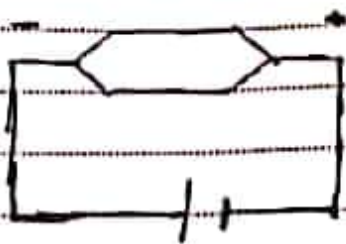
$\times / \div \Rightarrow$ the least number of s.f.

$+ / - \Rightarrow$ the least number of s.f. AFTER the

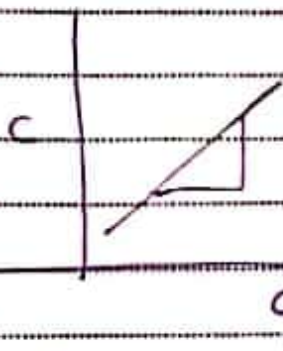
When a number without dot $\uparrow$ the answer should [dot]

Be the same

Subject: Radio activity



Electron charge = $1.6 \times 10^{-19} \text{ C}$
 mass = $9.1 \times 10^{-31} \text{ kg}$



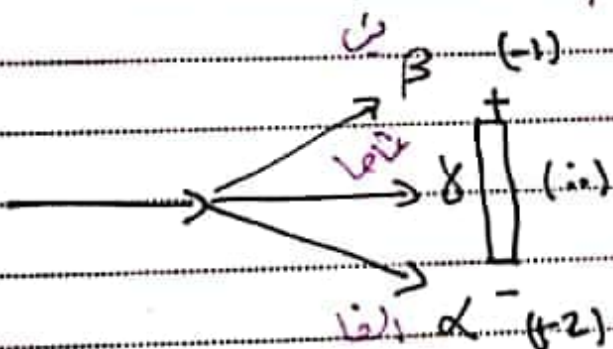
Electron mass = $\frac{e \cdot \text{charge}}{\text{Thompson charge to mass ratio}}$

$9.1 \times 10^{-31} \text{ kg}$

H2O_GROUP

Slope = Thompson concept.

[rays radioactive] α β γ



above
 [Gamma rays] a high energy radioactive it doesn't consist as [part

Rays mass ALMOST
 zero But not zero

particles have a mass
 it's too small compare
 to its energy

p + # n mass # 16

-2
 O
 electrons (16)
 8 + 2
 1

Atomic # 8

of protons + # of electrons (uncharged)

a # S T A R S N O T E B O O K

Subject:

Molar mass:

14.7

A_u

7.9

electrons:- 76

protons:- 79

neutrons:- 113

and so on...

H₂O GROUP

H: 1

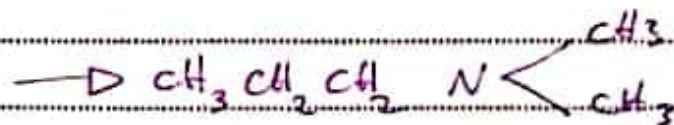
C: 12

O: 16

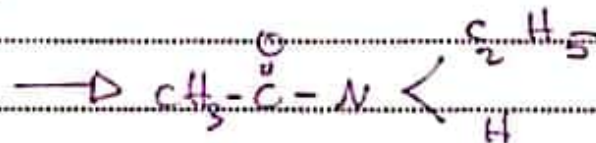
N: 14

S: 32

$$M = \frac{\text{mass}}{\text{molar mass}}$$

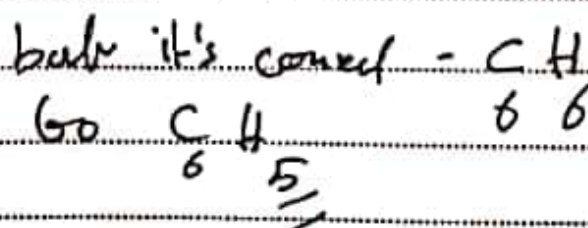
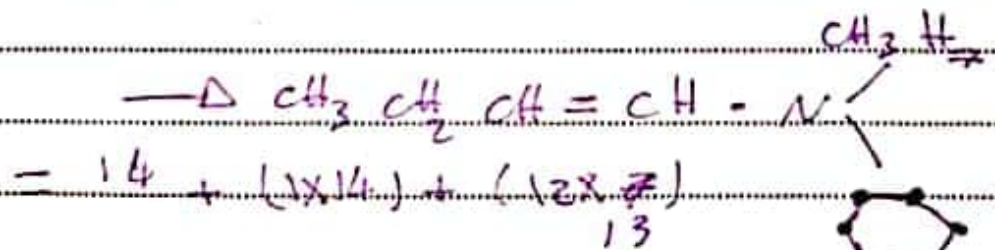


$$M_{\text{CH}_3} = 87 = 14 + (1 \times 13) + (1 \times 5)$$



$$= 14 + 16 + (1 \times 9) + (1 \times 4)$$

⇒ we don't consider the bonds - we're taking whole Atom.



Subject: Periodic Table

mass number = molar mass

Isotopes



Natural chloride is 75.8 (^{35}Cl) and chlorine isotope (^{37}Cl) is 24.2. What is the average weight of chlorine?

$$\text{Average mass} = \frac{75.8}{100} \times 35 + \frac{24.2}{100} \times 37$$

$$\frac{26.63}{100} + \frac{8.954}{100} = \frac{35.584}{100} = 35.5$$

Average Cl

Group $\longleftrightarrow$ Period

Metal :- electrically conductive / Thermally conductive
→ ductile
→ shiny appearance
→ Solid.

Non metals :- The opposite

1A $\rightarrow$ Alkali metal

2A $\rightarrow$ Alkali earth metal

6A $\rightarrow$ Chalcogens

7A $\rightarrow$ Halogens

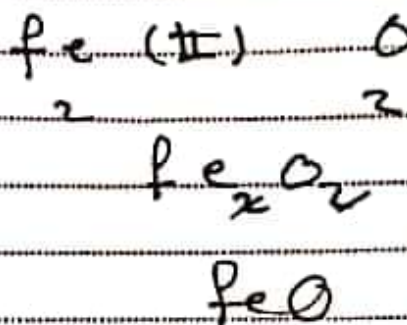
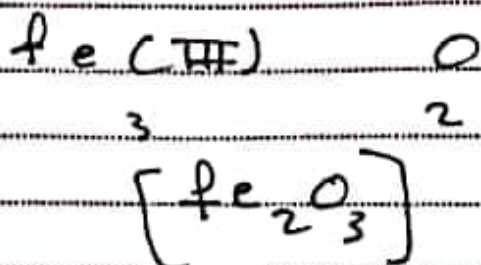
8A $\rightarrow$ Noble

H2O GROUP

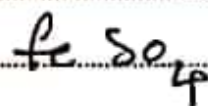
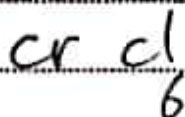
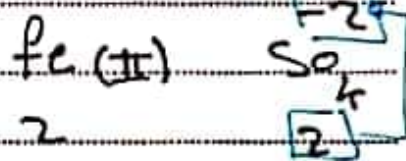
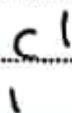
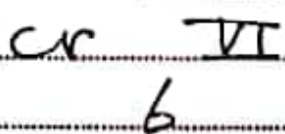
Valence e⁻ 1 2 3 4 3 2 1

► Subject: Stoichiometry

Transition Fe (II) > Oxidation
(III) state



► With Transition element should be give the oxidation state.



H2O GROUP

→ How many moles are there in 10g (NaOH)

$$\text{D.M.M. } 1 + 16 + 32 = 49$$

$$\text{Na : 23}$$

$$\text{O : 16}$$

$$\text{H : 1}$$

$$\text{[2] \# m} = \frac{m}{\text{I.M.M.}} = \frac{10}{49} = 0.204$$

→ How many mg are there in 1.5×10^{-4} moles of CH_3COOH ?

Subject: Stoichiometry / chemical equations

$$\# = \frac{g}{60}$$

$$mm = 4 + 24 + 32 = \underline{60}$$

$$1.5 \times 10^{-4} = \frac{g}{60}$$

$$m = 9 \times 10^{-3} g$$

$$9 \times 10^{-3} \times 10^3 = \underline{9}$$

$$1 \text{ mol} \rightarrow 6.02 \times 10^{23}$$

→ How many chloride ions (Cl^-) are there in 76 mg CrCl_2

$$n = \frac{76 \times 10^{-3}}{111} = 0.68 \times 10^{-3}$$

$$\text{Cr} : 40$$

$$\text{Cl} : 35.5$$

$$mm = 111$$

$$1 \text{ m} \rightarrow 6.02 \times 10^{23}$$

$$0.68 \rightarrow X$$

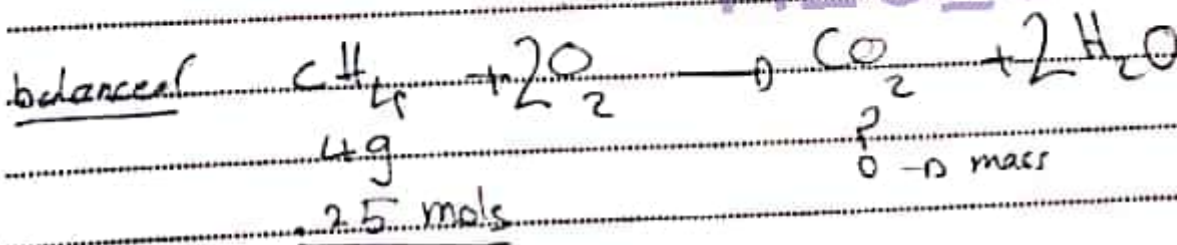
$$X = (4.09 \times 10^{20}) \times 2 = 8.18 \times 10^{20}$$

not finished yet

$$\text{Cr} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Chemical equations:-

H2O_GROUP



$$4g$$

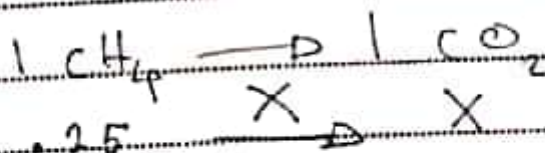
$$.25 \text{ mols}$$

$$\text{O} \rightarrow \text{mass}$$

$$\text{C} = 12$$

$$\text{O} = 16$$

$$\text{H} = 1$$



$$.25$$

$$\# = \frac{m}{mm}$$

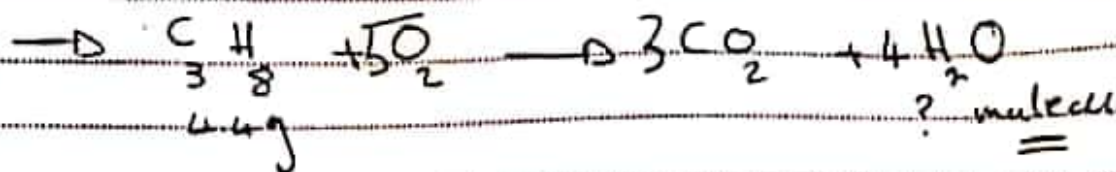
$$.25 = \frac{m}{44}$$

$$X = .25 \text{ mol (CO}_2\text{)}$$

mass

$$\underline{11}$$

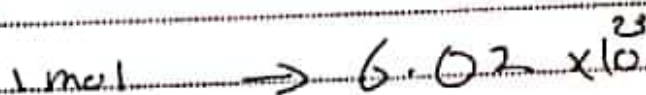
► Subject :



0.1 mol



$$X = 0.4 \text{ mol } \underline{\text{H}_2\text{O}}$$

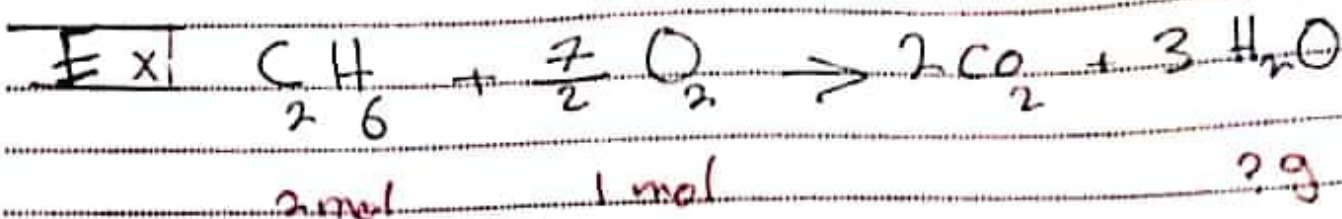


H2O_GROUP

$$\frac{7}{2} X = 0.2 \quad X = 0.057 \text{ moles}$$

$$0.057 = \frac{m}{44} \quad m = 2.5 \text{ g}$$

Subject: Limiting Reactant

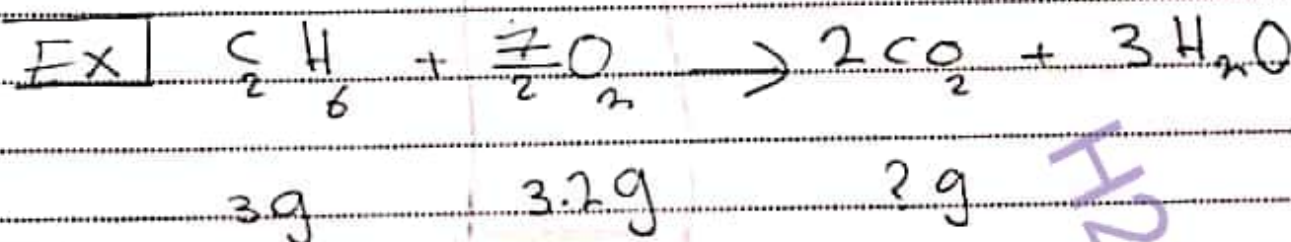


Unitless $\frac{2}{1} = \boxed{2}$ $\frac{1}{\frac{7}{2}} = \boxed{0.285}$

it's the least
so we use
it in calculations

$\frac{7}{2} \rightarrow 3$
 $1 \rightarrow x$

$\frac{7}{2} x = 3 \Rightarrow x = 0.8 \text{ mol H}_2\text{O} = \frac{m}{M} = \frac{m}{18} = \boxed{}$



II) to moles

$\frac{3}{30} = \boxed{0.1}$ $\frac{3.2}{32} = \boxed{0.1}$

III) decided on less

$\frac{0.1}{1} = \boxed{0.1}$ $\frac{0.1}{0.35} = \boxed{0.285}$
less $\ll$

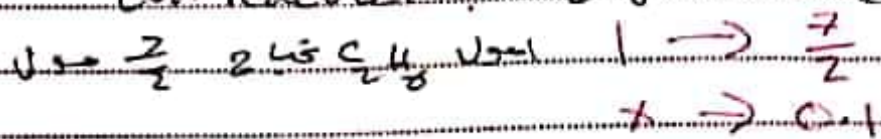
$\frac{7}{2} \rightarrow 0.2 \quad x = 0.057 \text{ moles CO}_2$ continued

Cal $\rightarrow x \quad 0.057 = \frac{m}{44} = 2.5 \text{ g}$
Fractional $\rightarrow$

► Subject: Unreacted / % yield

B) How many grams of C_2H_6 were left

Unreacted? C_2H_6 O_2



$x = 0.0285 \text{ mol (Reacted)}$

Unreacted = (Total - reacted)

$$= 0.1 - 0.0285$$

$$\text{excess} = 0.0715 \text{ mol unreacted} = \frac{m}{M} = \dots$$

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{Theoretical yield}} \times 100$$

C) if .1 g CO_2 were produced By experiment
[actual yield] What's the percent yield of CO_2

$$\% \text{ yield} = \frac{.1 \text{ g}}{2.5 \text{ g}} \times 100\% = \underline{4\%}$$

BAD

to be more accurate it should be $> 90\%$

H2O_GROUP good $< 90\%$

Subject: Empirical / molecular formula

الاصغر → Empirical formula → the simplest form

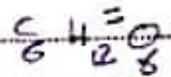
الكبير ← Molecular formula → we can simplify it

molecular = $[n]$ (empirical for)
calculable

$$\frac{C \ H \ O}{6 \ 12 \ 6} = \frac{180}{30} = 6$$

ex: $6(C_2H_2O_2)$

القيمة الجزيئية
القيمة المولية



Ex: C 80% H 20%

The total is 100%

What's the empirical formula?

What's the molecular formula?

$$\frac{C}{6.6} = \frac{80}{12}$$

$$\frac{H}{20} = \frac{20}{1}$$

100%
mm

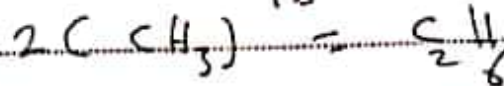
$$\frac{1}{6.6} : \frac{3}{20} = \frac{20}{6.6}$$

نسبة على اجزاء
مول

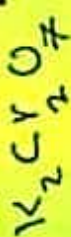


given

CH₃ the whole number $\frac{30}{10} = 3$



Empirical formula
molecular formula



النسبة
مباشرة مع الجزيء

H2O_GROUP

► Subject: Percentage composition



$$\%C = \frac{2 \times 12}{30} \times 100\% = 80\%$$

$$\%H = \frac{6 \times 1}{30} \times 100\% = 20\%$$

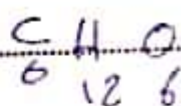
$$\% \times = \frac{\# \text{ of atom (x)} \times \text{elm (x)}}{\text{Mm compound}}$$

$$\# \text{ Molecules} = \# \text{ moles} \times N.A$$

$$\# \text{ Atoms} = \# \text{ moles} \times N.A \times \# \text{ atom}$$

avogadro's number $10^{23} \times 0.62 \leftarrow$

Ex



$$\frac{180}{180}$$

$$\text{Mm} = 180$$

molecules

1. calculate # of moles ?

2. " " " " (H) ?

3. " " " " (C) ?

$$\textcircled{1} = \frac{18}{180} \times 6.022 \times 10^{23} = 6.02 \times 10^{23} \text{ molecules}$$

$$\textcircled{2} =$$

H2O_GROUP